

Mohd Bilal Khan

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Summary

Final-year ECE undergraduate at Indian Institute of Information Technology, Pune (graduating 2027) with strong backend engineering fundamentals, built through production-grade internship experience. Expertise spans full-stack development (FastAPI, React, Node.js), Generative AI applications, and scalable data pipelines. Ranked Top 1.3% at HackRx among 10,000+ teams and solved 500+ DSA problems across CodeChef, Codeforces, and LeetCode.

Education

Indian Institute of Information Technology, Pune (CGPA: 7.40)

Pune, India

Bachelor of Technology in Electronics and Communication Engineering

August 2023 – May 2027

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Digital Logic Design, Embedded Systems, Microprocessors, Object-Oriented Programming, Database Management

Experience

Digital Forensics Lab, IIIT Pune

Pune, India

Research Intern (Team Lead)

Oct 2025 – June 2026

- Analyzed Autopsy's forensic workflows and architected **Sentinel Core**, a GenAI-native full-stack digital evidence triage platform (React/TypeScript, FastAPI) that uses **LLMs** to cross-reference extracted disk and storage evidence against criminal record data; led a 4-person team on this INR 4.2 crore government-funded initiative (Forensics Department of India).
- Shipped multimodal detection pipelines — YOLOv8 + CLIP for image threats, BERT-NER/BART for text and PII, Whisper for audio, and a FAISS-indexed SigLIP deepfake detector — unified under an explainable, LLM-scored evidence clustering layer.

Technical Skills

Programming Languages: C++, Go, Python, JavaScript, TypeScript, C

Frameworks & Libraries: Node.js, React.js, Express.js, FastAPI, LangChain, LangGraph

Generative AI & NLP: LLM Agents, RAG, Vector Embeddings, Prompt Engineering, Gemini

Database Systems: MongoDB, Supabase, MySQL

Version Control & Tools: Docker, Git, GitHub

Projects

Sanjaya-AI, AI-Powered Multi-Agent Research Platform [sanjaya-ai.onrender.com](#) | [Github](#)

- Designed a master-agent orchestration system using **LangGraph** in Python and FastAPI, dynamically routing queries to specialized research agents across multiple domains.
- Leveraged **Gemini 2.5 Flash** via the OpenAI SDK for query understanding, agent selection, and structured output generation; built end-to-end PDF report pipelines using ReportLab with automated REST-triggered workflows.
- Integrated Supabase, public REST APIs, internal document parsing, and real-time web intelligence into a unified research pipeline with normalized schemas and data validation; developed a responsive React/Vite/TypeScript/Tailwind frontend.

Le Rituel, AI-Powered Skincare Platform [nodeflowai.in](#) | [Github](#)

- Built **Le Rituel** ([nodeflowai.in](#)), a production-deployed AI skincare platform that integrates **Claude Sonnet 4.5** (vision + chat) to analyse selfies for skin-type detection and power a context-aware, routine-personalized chat assistant — with photos discarded server-side post-inference for privacy.
- Engineered a secure, containerised full-stack system (**FastAPI** + **React 19** + **MongoDB**) with a score-based routine recommendation engine, **JWT auth** (15-min access / 7-day refresh, JTI revocation blacklist), **Fernet encryption** for sensitive fields at rest, brute-force lockout, and a three-container **Docker Compose** deployment behind nginx reverse proxy.

Achievements

- Secured Rank 130 (Top 1.3%) out of 10,000+ teams in the HackRx national hackathon, and advanced to the finals at the IIT Roorkee Hackathon.
- Solved 500+ problems across data structures, graph theory, and DP, achieving a max rating of 1521 on CodeChef and 1221 on Codeforces.
- Achieved a Top 2% rank among 1.1 million candidates in the JEE Mains examination.

Extracurriculars

- Google DSC Campus Ambassador, IIIT Pune** — Led community initiatives and fostered technical growth among university peers; also an active open-source **Contributor at GirlScript Summer of Code** (2025).
- Athlete, Inter-IIIT National Football Tournament** — Represented the university and won Silver at the collegiate-level national meet.